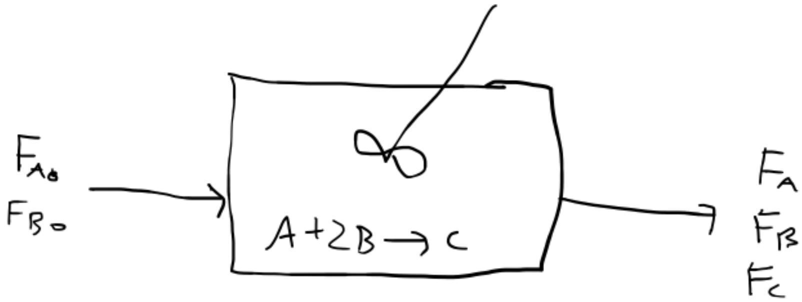
E01 Practicing with Stoichiometric Tables and Changes in Volumetric Flowrate

The gas phase reaction A + 2B --> C is carried out isothermally and isobarically in a fluidized CSTR with no pressure drop. The feed is stoichiometric in A and B and no C is fed. The pressure, temperature, and volumetric flowrate of the feed gas are 6 atm, 170°C, and 50 L/min respectively. The rate constant k_A = 53 mol/kg catalyst*min*atm³ at 300 K, and the activation energy is 80 kJ/mol. Complete the stoichiometric table and write the rate as a function of conversion.



Species	initial	change	exiting
A	$F_{A0} = \frac{y_{A0} P_0 v_0}{R T_0}$	$-F_{A0} X_A$	$F_A = F_{A0}(1-X_A)$
B	$F_{B0} = 2F_{A0}$	$-F_{B0} X_B = -2F_{A0} X_A$	$F_B = 2F_A = 2F_{A0}(1-X_A)$
C	0	$F_{A0} X_A$	$F_C = F_{A0} X_A$

$$F_{A0} = \frac{(\frac{1}{3})(6\,atm)(50\,\frac{L}{min})}{(0.0821\,\frac{L\,atm}{mol\,K})(170°C + 273.15\,K)}$$

$$y_{A0} = \frac{1\,mol\,A}{1\,mol\,A + 2\,mol\,B} = \frac{1}{3}$$

- assume elementary kinetics

$-r_A = k_A C_A C_B^2$. for fluidized CSTR, the units on the rate should be equal to $\frac{mol}{time \cdot mass\ catalyst}$. This is because the mole balance for a fluidized CSTR is $W = \frac{F_{A0} X_A}{-r_A}$ instead of $V = \frac{F_{A0} X_A}{-r_A}$ for a "normal" CSTR.

- so rate is: $-r_A = k_A P_A P_B^2$. Recalling that

$$C_i = \frac{P_i}{RT} = \frac{y_i P_{tot}}{RT}$$

$$-r_A = k_A P_A P_B^2$$

Eq 4-25 or Eq 4-27

$$P_A(X_A) = RT\, C_A(X_A)$$

$$P_i = \frac{P_{A0}(\Theta_i + \nu_i X_A)}{1+\varepsilon X_A} P$$

P here = 1 since the reactor is isobaric

$$\varepsilon = y_{A0}\,\delta \quad , \quad \delta = \left(\frac{d}{a} + \frac{c}{a} - \frac{b}{a} - 1\right) = -2$$

$$\cancel{} \quad \frac{1}{1} - \frac{2}{1} - 1$$

$$\varepsilon = -\frac{2}{3}$$

$$P_A = \frac{P_{A0}(1 - X_A)}{1 - \frac{2X_A}{3}} \quad , \quad P_B = 2P_A$$

$$P_{A0} = y_{A0}\,P_0$$